Carburising flame

This flame is also known as reducing flame. The supply of acetylene will be more than oxygen to produce this flame. Carburising flame consists of three zones namely,

1. Sharp inner cone
2. White intermediate cone
3. Bluish outer cone

Carburising flame is useful in welding low carbon steel and alloy steels. It is also used to harden the outer surface of metal parts.

Oxidising flame

Oxidising flame is obtained by supplying more oxygen than acetylene. It consists of two zones namely bright inner cone and outer cone. This flame is useful in welding brass and bronze.

7.13 Filler rods used in gas welding

Filler rods used in gas welding supply the additional metal is making joints. These rods are melted by the gas flame and deposited over the parts of the joint. Generally the filler rods are made of the same metal as that of the parts of the joint.

The diameter of the filler rod depends upon the thickness of the parts to be welded. The strength of the welding joint is increased by adding Nickel or Chromium in filler rods. A thin coat of copper is provided on the filler rods to prevent the molten metal from reacting with atmospheric oxygen. Flux may be applied either in powdered form or liquid form.

7.14 Advantages of gas welding

1. Applied for different classes of work
2. Welding temperature is controlled easily
3. The quantity of filler metal added in the joint can easily be controlled
4. The cost of the welding unit is less
5. The cost of maintenance is less
6. Both welding and cutting can be done

7.15 Disadvantages of gas welding

1. Intended for welding thin work pieces only
2. The process of welding is slow
3. The time taken by the gas flame to heat the metal is more when compared with electric arc.
4. The strength of the joint is less
5. Great care should be taken in handling and storing gas cylinders.

7.16 Differences between arc welding and gas welding

Arc welding	Gas welding
The heat is generated by the electric arc	The required heat is provided by the gas flame.
The working temperature is about 4000^0C	The temperature of the gas flame is about 3200^0C
The filler rod is used as electrode.	Filler rod is used separately
Care should be taken against the dangers that may be caused because of electricity	The danger of the process is because of the gases at high pressure

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